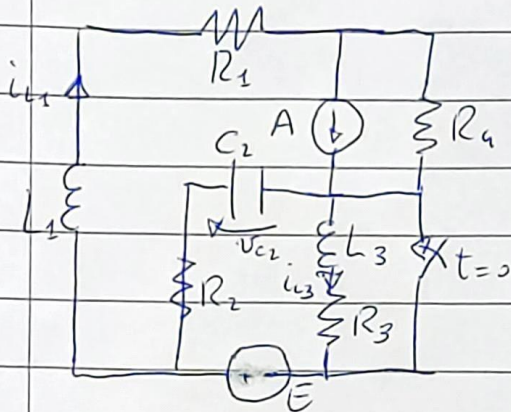


TRANSITORIO



$$E = 2V$$

$$R_4 = 3\Omega$$

$$A = 3A$$

$$L_1 = 2mH$$

$$L_3 = 3mH$$

$$C_2 = 1mF$$

$$R_1 = 2\Omega$$

$$R_2 = 2\Omega$$

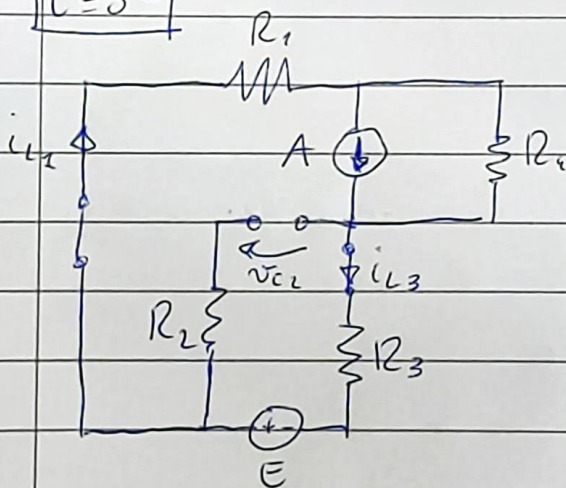
$$R_3 = 1\Omega$$

$$t=0^-: W_{L1}, W_{L3}$$

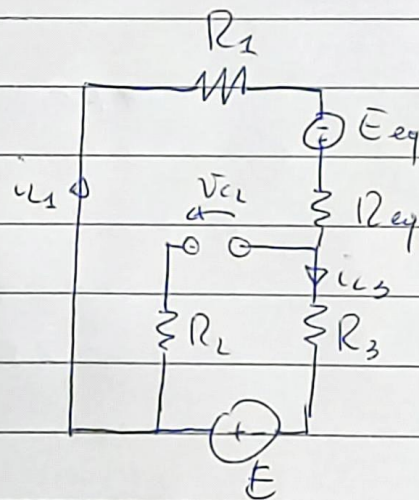
$$t=0^+: \frac{di_{L1}}{dt}, \frac{di_{L3}}{dt}$$

$$t=\infty: P_2, Q_{C2}, P_{R3}$$

$$t=0$$



→



$$E_{eq} = A R_4 = 9V$$

$$R_{eq} = R_4 = 3\Omega$$

$$i_{L1} = i_{L3}$$

$$i_{L1} = i_{L3} = \frac{E_{eq} + E}{R_1 + R_{eq} + R_3} = 1.83A$$

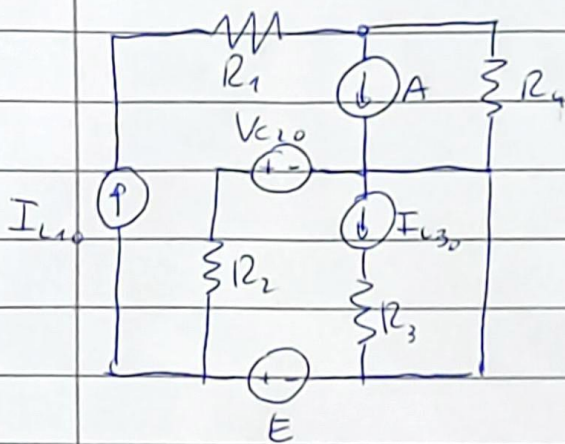
$$W_{L1} = \frac{1}{2} L_1 i_{L1}^2 = 3.35mJ$$

$$W_{L3} = \frac{1}{2} L_3 i_{L3}^2 = 5.02mJ$$

$$KUT: v_{C2} - E + v_{R3} = 0 \rightarrow v_{C2} = E - v_{R3} = E - i_{L1} R_3 = +0.17V$$

$$W_{C2} = \frac{1}{2} C_2 v_{C2}^2 = 0.014mJ$$

$t = 0^+$



$$v_{L1}(t=0^-) = v_{L1}(t=0^+) = V_{L10}$$

$$i_{L1}(t=0^-) = i_{L1}(t=0^+) = I_{L10}$$

$$i_{L3}(t=0^-) = i_{L3}(t=0^+) = I_{L30}$$

$$\text{KUC: } I_{L10} - A - i_{R4} = 0$$

$$i_{R4} = I_{L10} - A = -1.17 \text{ A}$$

$$v_{R4} = i_{R4} R_4 = -3.51 \text{ V}$$

$$\text{LKT: } v_{L1} - E + v_{R4} + v_{R1} = 0 \rightarrow$$

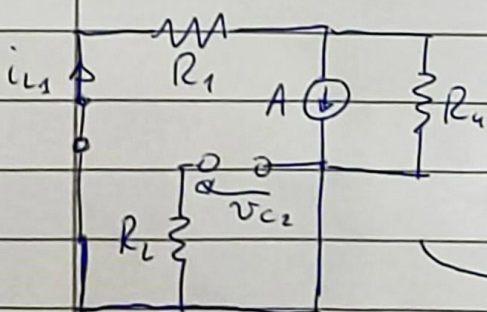
$$\rightarrow v_{L1} = E - v_{R4} - v_{R1} = E - v_{R4} - R_1 I_{L10} = 1.85 \text{ V}$$

$$\frac{di_{L1}}{dt} = \frac{v_{L1}}{L_1} = 925 \text{ A/s}$$

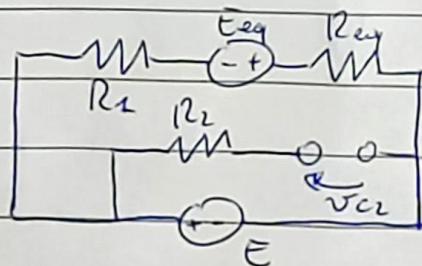
$$\text{LKT: } v_{L3} = -v_{R3} = -R_3 I_{L30} = -1.83 \text{ V}$$

$$\frac{di_{L3}}{dt} = \frac{v_{L3}}{L_3} = -610 \text{ A/s}$$

$t = \infty$



$$P_{R3} = i_{R3} R_3 = 0 \text{ W}$$



$$E_{eq} = A R_4 = 9 \text{ V}$$

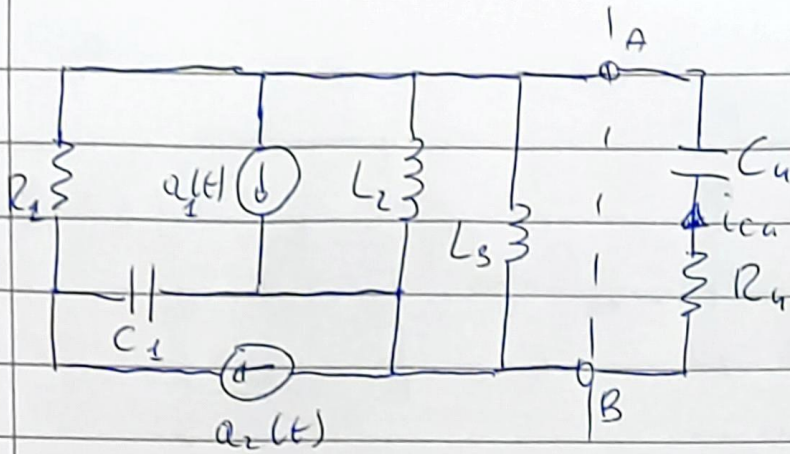
$$R_{eq} = 3 \Omega$$

$$i_E = \frac{E_{eq} + E}{R_1 + R_{eq}} = 2.2 \text{ A}$$

$$P_E = i_E E = 4.4 \text{ W}$$

$$\text{LKT: } E - v_{C2} = 0 \quad v_{C2} = E \quad Q_{C2} = v_{C2} C_2 = 2 \text{ mC}$$

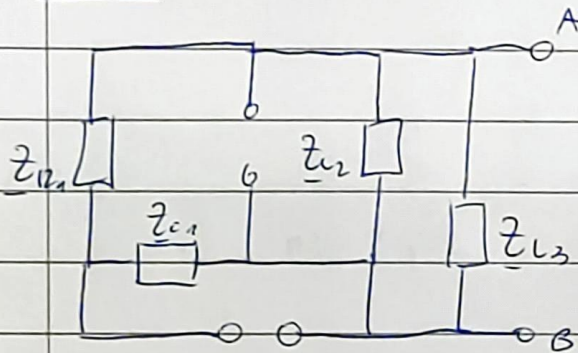
REGIME SINUSOÏDALE - THEVENIN



$$\begin{aligned} R_1 &= 2 \Omega & L_3 &= 2 \text{ mH} \\ R_4 &= 4 \Omega & L_2 &= 1 \text{ mH} \\ C_1 &= 2 \text{ mF} & C_4 &= 4 \text{ mF} \\ a_1(t) &= 2 \cos(2t + 15^\circ) \text{ A} \\ a_2(t) &= 3 \sin(2t - 15^\circ) \text{ A} \\ &= 3 \cos(2t - 105^\circ) \text{ A} \end{aligned}$$

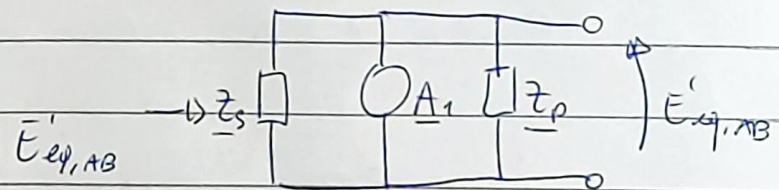
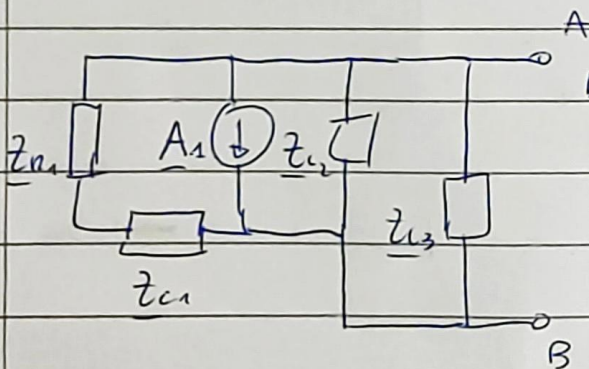
$$\begin{aligned} \underline{z}_{R1} &= 2 \Omega, \quad \underline{z}_{R4} = 4 \Omega, \quad \underline{z}_{C1} = -\frac{1}{j\omega C_1} = -250 j \Omega, \\ \underline{z}_{L2} &= j\omega L_2 = 0.002 j \Omega, \quad \underline{z}_{L3} = 0.004 j \Omega, \quad \underline{z}_{C4} = -125 j \Omega \\ \underline{A}_1 &= \frac{1}{\sqrt{2}} \angle 15^\circ \text{ A}, \quad \underline{A}_2 = \frac{3}{\sqrt{2}} \angle -105^\circ \text{ A} \end{aligned}$$

$\underline{z}_{eq}$



$$\begin{aligned} \underline{z}_{eq} &= ((\underline{z}_{R1} + \underline{z}_{C1}) \parallel \underline{z}_{L2}) \parallel \underline{z}_{L3} \\ &= 0.001 j \Omega \end{aligned}$$

$\underline{A}_1 \neq 0$



$$\underline{z}_s = \underline{z}_{R1} + \underline{z}_{C1} = 2 - 250 j \Omega$$

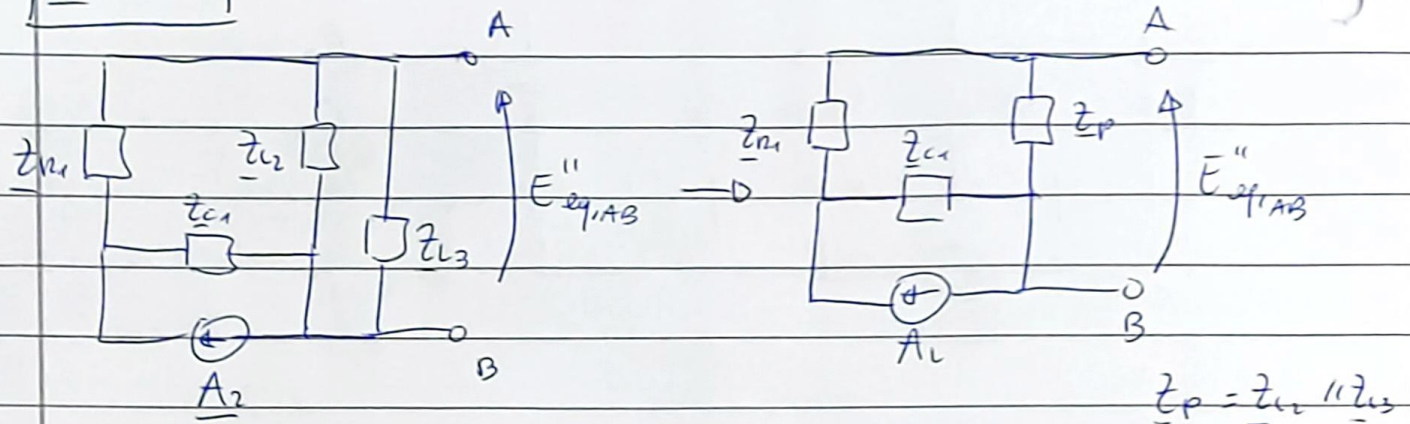
$$\underline{z}_p = \underline{z}_{L2} \parallel \underline{z}_{L3} = 0.001 j \Omega$$

$$\underline{E}'_{eq,AB} = -\underline{A}_1 \frac{\underline{z}_s}{\underline{z}_s + \underline{z}_p} \underline{z}_p =$$

$$= 0.0005 - 0.002 j \text{ V}$$

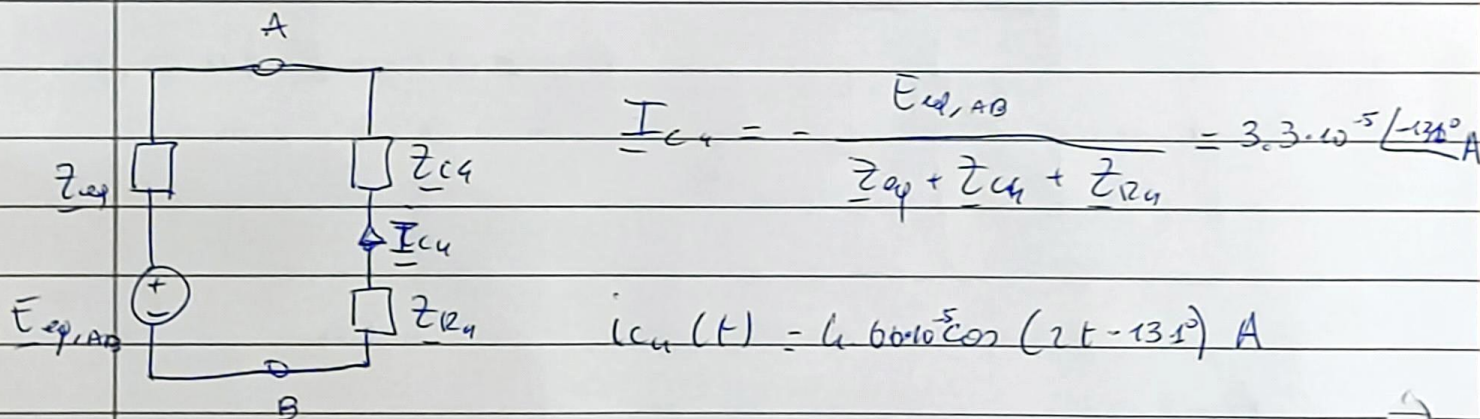
$$= 0.002 \angle -75^\circ \text{ V}$$

$$\underline{A}_2 \neq 0$$



$$\underline{E}''_{eq,AB} = \underline{A}_2 \frac{z_{c1}}{z_{n1} + z_p + z_{c1}} \quad \underline{z}_p = 0.003 - 0.001j \Omega = 0.003 \angle -15^\circ \Omega$$

$$\underline{E}_{eq,AB} = \underline{E}'_{eq,AB} + \underline{E}''_{eq,AB} = 0.003 - 0.003j \text{ V} = 0.004 \angle -39^\circ \text{ V}$$



POTENZA ASSORBITA C_4

$$P_{C4} = 0 \quad Q_{C4} = -\frac{1}{\omega C_4} |\underline{I}_{C4}|^2 = -1.36 \cdot 10^{-7} \text{ VAR}$$